MX

In [ ]:
```
#%dhist
#0: PT3S\nb (notebooks ipynbDir)
%cd -0
%cd ..
# notebooks workingDir should be PT3S
```

In [ ]:
```
!pip uninstall PT3S -y
```

In [3]:
```
!pip install PT3S
```

```
Collecting PT3S
  Using cached PT3S-0.0.16.dev1-py3-none-any.whl
Installing collected packages: PT3S
Successfully installed PT3S-0.0.16.dev1
```

In [4]:
```
from PT3S import Mx
```

In [5]:
```
mx=Mx.Mx('./testdata/M-1-0-1.MX1'
         ,unpackVectorChannels=False
         ,channelsNotToBeProcessedSir3sIDRegExp=[
                         '(\S+)~(\S+)~(\S+)~(\S+)~*INST*'
                        ,'^ROHR_VRTX' # surprise: not a VectorChannel?!
                        ,'^ROHR~(\S+)~(\S+)~(\S*)~QM[I,K]{1}'
                        ,'^VENT~(\S+)~(\S+)~(\S*)~OEFFNET'
                        ,'^VENT~(\S+)~(\S+)~(\S*)~SCHLIESST'
                        ,'^VENT~(\S+)~(\S+)~(\S*)~AUF'
                        ,'^VENT~(\S+)~(\S+)~(\S*)~ZU'
                        ,'^KNOT~(\S+)~(\S*)~(\S*)~T'
                        ,'^KNOT~(\S+)~(\S*)~(\S*)~PDAMPF'
                        ,'^KNOT~(\S+)~(\S*)~(\S*)~RHO'
                        ,'^PUMP'
                        ,'^PGRP'
                        ,'^R\S+~(\S+)~(\S*)~(\S*)~X\S+'
                      ])
```

to Pandas

MX-Channel TIMESTAMP (1st MX-Channel) used as DataFrame Index; all other MX-Channels are DataFrame Columns

In [6]:
```
import pandas as pd
```

getting a DataFrame

In [7]:
```
df=mx.setResultsToMxsZipfile('./testdata/M-1-0-1.ZIP')
```

In [8]:
```
type(df)
```

Out[8]:
```
pandas.core.frame.DataFrame
```

in Pandas

Series

In [9]:
```
ts=df.iloc[1:-2,0]
type(ts)
```

Out[9]:
```
pandas.core.series.Series
```

Timestamps

In [10]:
```
ts.index[0]
```

Out[10]:
```
Timestamp('2017-06-18 23:00:00')
```

In [11]:
```
ts.index[-1]
```

Out[11]:
```
Timestamp('2017-06-19 00:30:00')
```

DatetimeIndex

In [12]:
```
type(ts.index)
```

Out[12]:
```
pandas.tseries.index.DatetimeIndex
```

Time Zone (tz_convert(None) will remove timezone after converting to UTC time)

In [13]:
```
print(ts.index.tz)
```

```
None
```

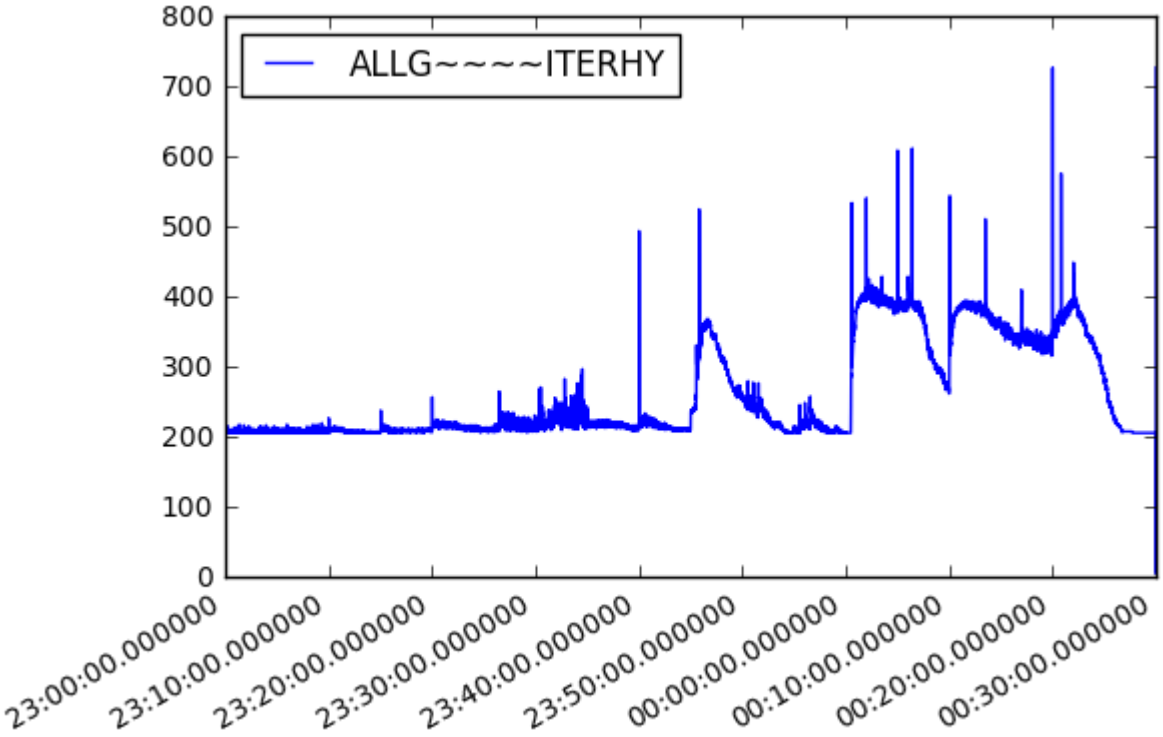
Time Series Plot

In [14]:
```
%matplotlib inline
import matplotlib.pyplot as plt
from matplotlib import dates
```

by MX-Channel-Idx

In [15]:
```
plt.figure()
df.iloc[:,17].plot()
plt.legend(loc='best')
```

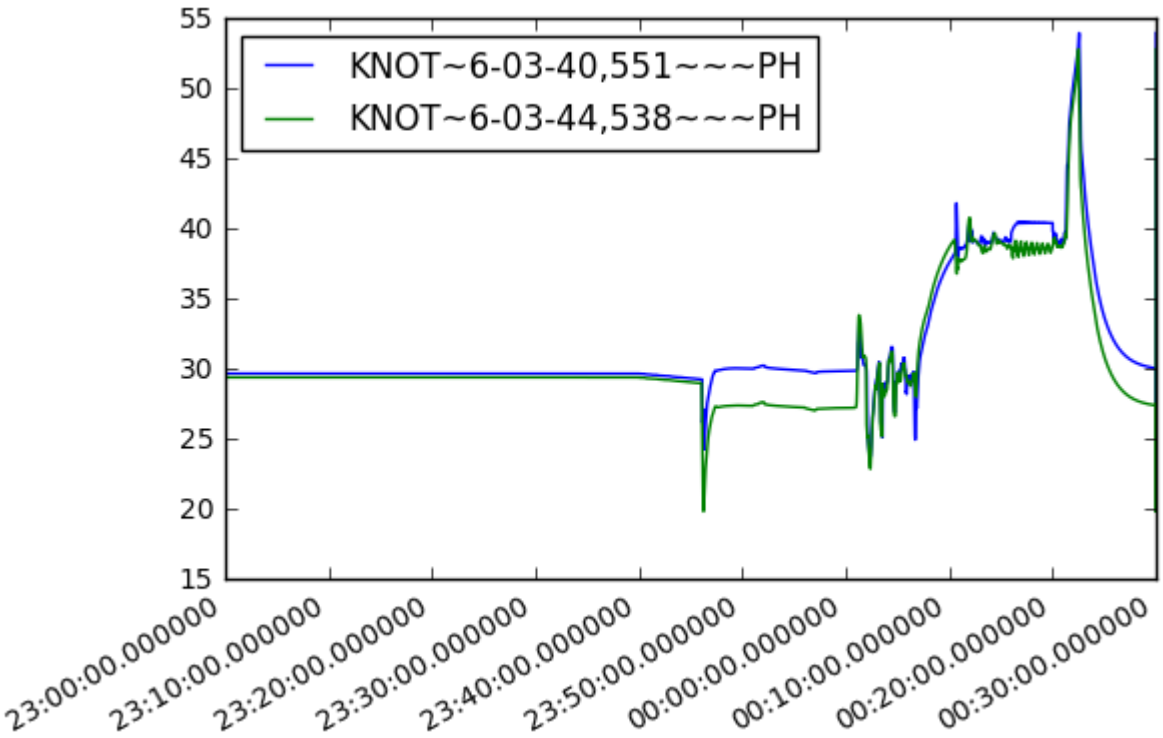
Out[15]: <matplotlib.legend.Legend at 0x19b322579e8>



by MX-Channel-Sir3sID

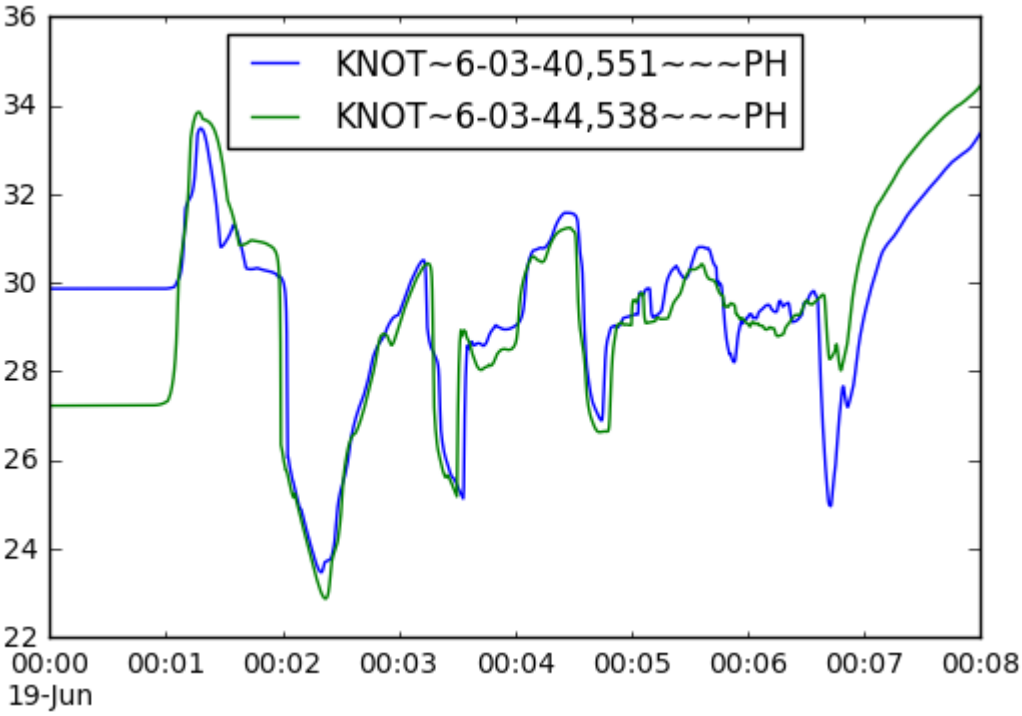
In [16]:
```
import re
regExpCompiled=re.compile(r'^KNOT~6-03-4\d+(\S+)~(\S*)~(\S*)~PH')
df_colsel=df.loc[:,sorted([column for column in df.columns if regExpCompiled.search(c
olumn) != None])]
df_colsel.plot()
```

Out[16]: <matplotlib.axes._subplots.AxesSubplot at 0x19b448397b8>



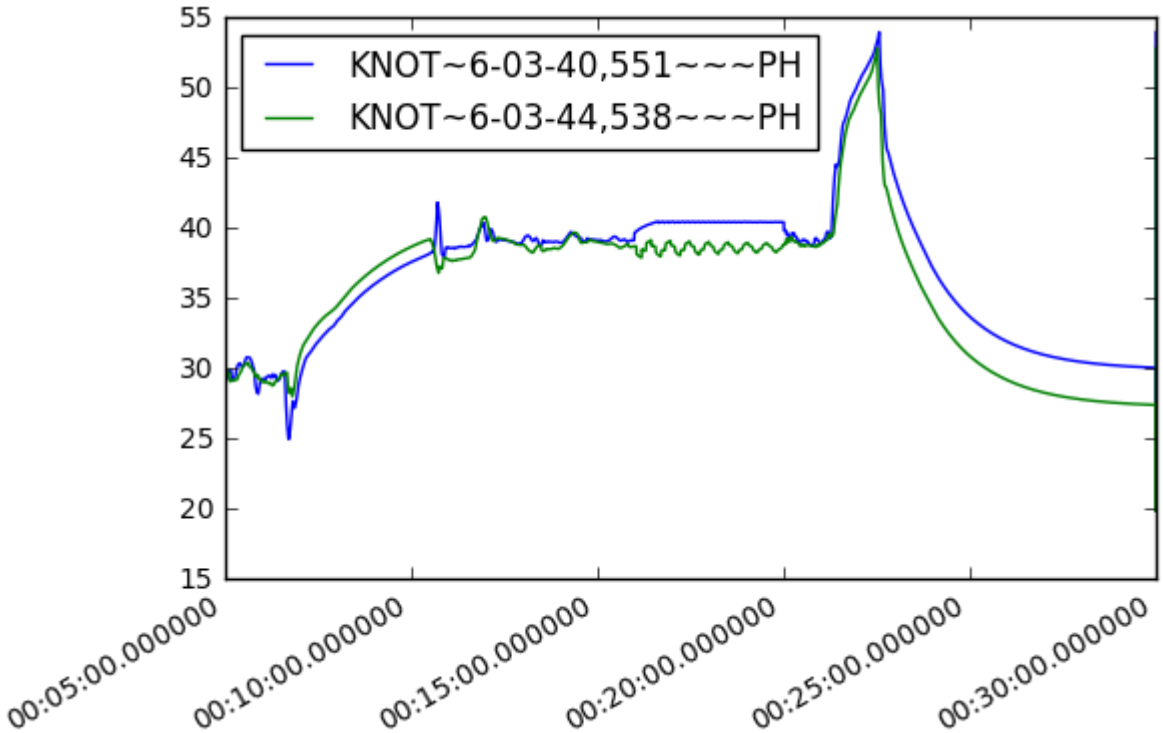
In [17]:
```
start=df_colsel.index[0].to_datetime()+pd.DateOffset(minutes=60)
end=start+pd.DateOffset(minutes=8)
df_colsel.between_time(start_time=start.time(), end_time=end.time()).plot()
```

Out[17]: <matplotlib.axes._subplots.AxesSubplot at 0x19b4484ba90>



In [18]:
```
df_colsel[pd.datetime(2017,6,19,0,5):].plot()
```

Out[18]: <matplotlib.axes._subplots.AxesSubplot at 0x19b3fb069e8>



SnapShotType

In [19]:
```
len(df)
```

Out[19]: 21604

In [20]:
```
df.loc[:,[r'ALLG~~~~SNAPSHOTTYPE']].head(1)
```

Out[20]:

	ALLG~~~~SNAPSHOTTYPE
2017-06-18 23:00:00	b'STAT'

In [21]:
```
df.loc[:,[r'ALLG~~~~SNAPSHOTTYPE']].tail(2)
```

Out[21]:

	ALLG~~~~SNAPSHOTTYPE
2017-06-19 00:30:00	b'TMIN'
2017-06-19 00:30:00	b'TMAX'

In [22]:
```
len(df[df[r'ALLG~~~~SNAPSHOTTYPE'].isin([b'STAT',b'TMIN',b'TMAX'])])
```

Out[22]: 3

In [23]:
```
df.loc[:,
[r'ALLG~~~~SNAPSHOTTYPE',r'ALLG~~~~ITERHY']].groupby(r'ALLG~~~~SNAPSHOTTYPE').count()
```

Out[23]:

	ALLG~~~~ITERHY
ALLG~~~~SNAPSHOTTYPE	
b'STAT'	1
b'TIME'	21601
b'TMAX'	1
b'TMIN'	1

In [24]:
```
df2=df[df[r'ALLG~~~~SNAPSHOTTYPE'].isin([b'TIME'])]
```

In [25]:
```
len(df2)
```

Out[25]: 21601

In [26]:
```
df2.loc[:,'KNOT~6-03-44,538~~~PH'].head(4)
```

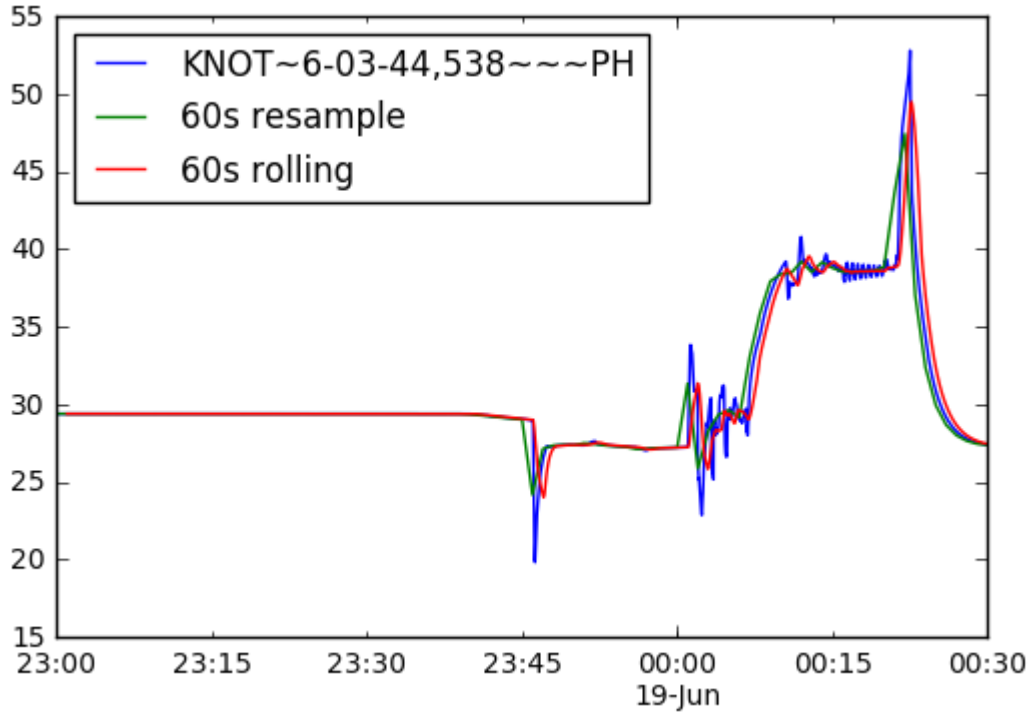
Out[26]:
```
2017-06-18 23:00:00.000    29.404100
2017-06-18 23:00:00.250    29.404099
2017-06-18 23:00:00.500    29.404100
2017-06-18 23:00:00.750    29.404100
Name: KNOT~6-03-44,538~~~PH, dtype: float64
```

Time Series Functions

In [27]:
```
ts=df2.loc[:,'KNOT~6-03-44,538~~~PH']
```

In [28]:
```
ts60=df2.loc[:,'KNOT~6-03-44,538~~~PH'].resample('60s').mean()
ts60.name='60s resample'
ts60r = ts.rolling(window=60*4).mean()
ts60r.name='60s rolling'
plt.figure()
ts.plot()
ts60.plot()
ts60r.plot()
plt.legend(loc='best')
```

Out[28]:
```
<matplotlib.legend.Legend at 0x19b47c0fe80>
```



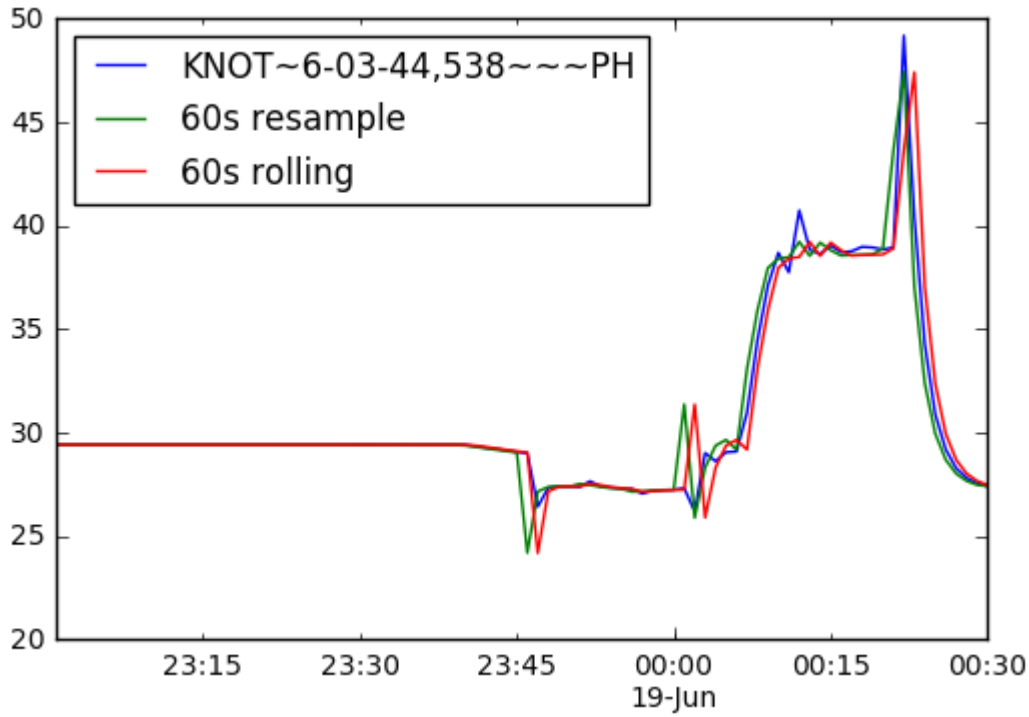
In [29]:
```
df3=pd.DataFrame({ts60.name : ts60,ts.name : ts,ts60r.name : ts60r})
df3=df3.reindex_axis([ts.name,ts60.name,ts60r.name], axis=1)
df3.head()
```

Out[29]:

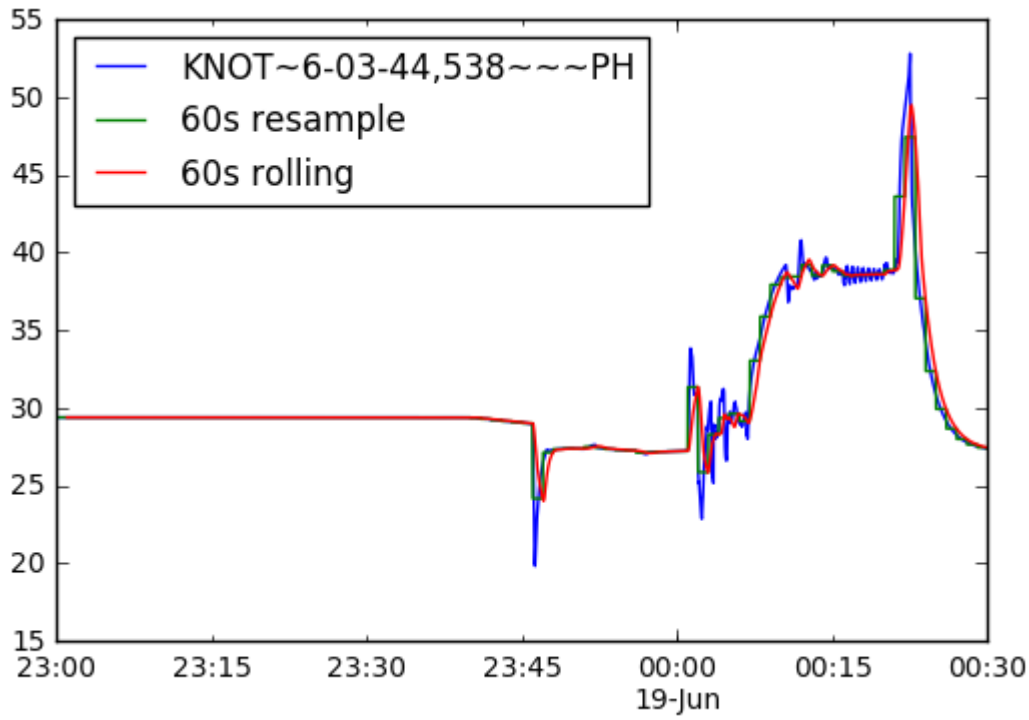
	KNOT~6-03-44,538~~~PH	60s resample	60s rolling
2017-06-18 23:00:00.000	29.404100	29.404088	NaN
2017-06-18 23:00:00.250	29.404099	NaN	NaN
2017-06-18 23:00:00.500	29.404100	NaN	NaN
2017-06-18 23:00:00.750	29.404100	NaN	NaN
2017-06-18 23:00:01.000	29.404099	NaN	NaN

In [30]:
```
df3.dropna().plot()
```

Out[30]: <matplotlib.axes._subplots.AxesSubplot at 0x19b4a53bc50>



In [31]:
```
pPlot = df3.ffill().plot()
```

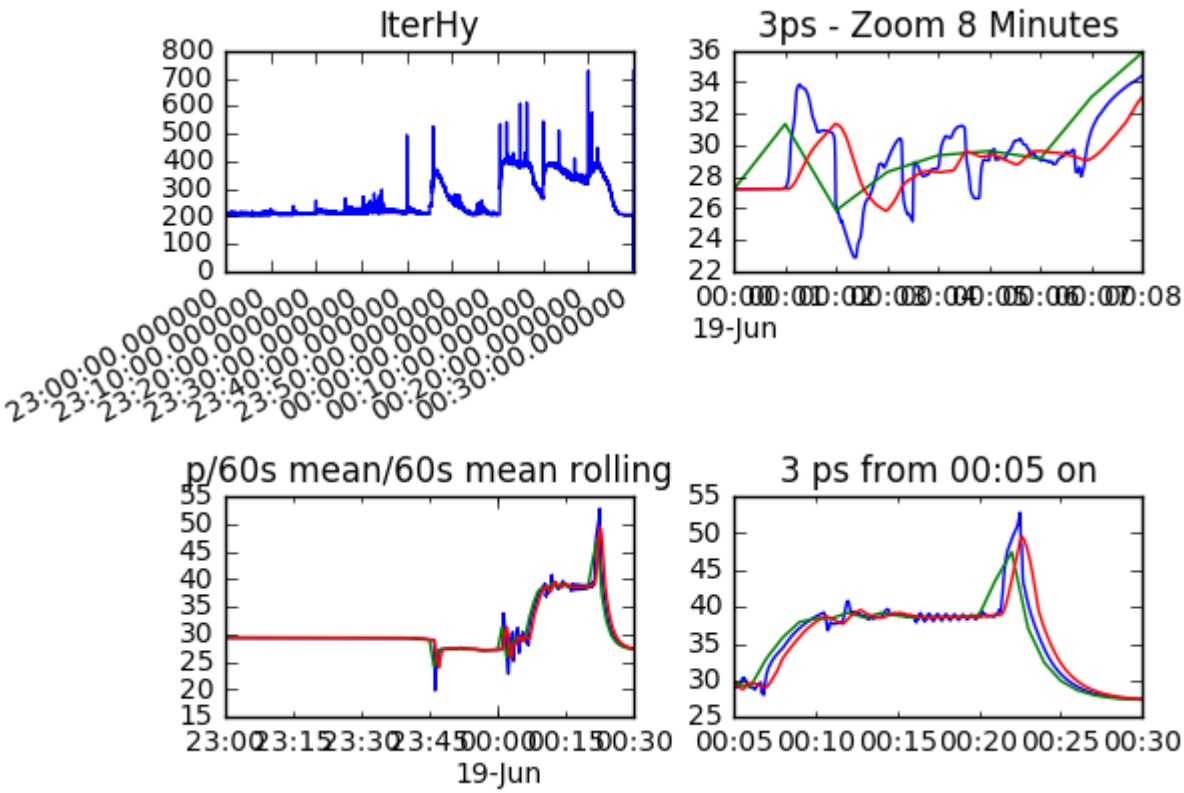


MATLAB, and pyplot, have the concept of the current figure and the current axes. All plotting commands apply to the current axes.

The function gca() returns the current axes (a matplotlib.axes.Axes instance), and gcf() returns the current figure (matplotlib.figure.Figure instance).

The subplot() command specifies numrows, numcols, num where num ranges from 1 to numrows*numcols.

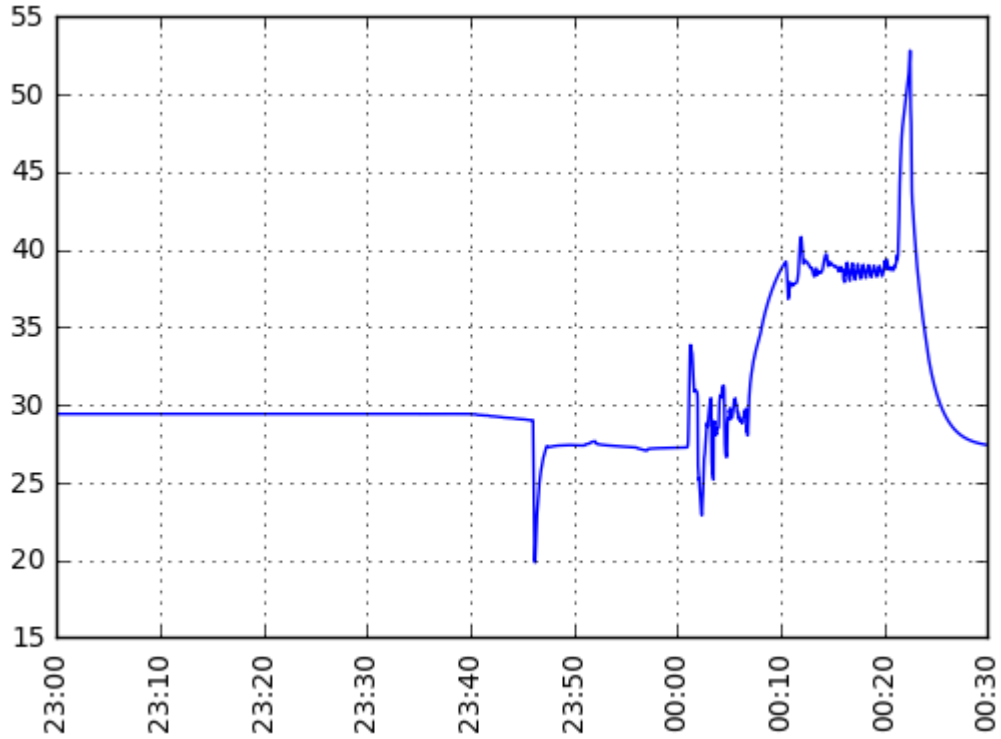
In [32]:
```
plt.figure(1)

plt.subplot(221)
plt.title('IterHy')
df.iloc[:,17].plot()

plt.subplot(223)
plt.title('p/60s mean/60s mean rolling')
ts.plot()
ts60.plot()
ts60r.plot()

plt.subplot(222)
plt.title('3ps - Zoom 8 Minutes')
ts.between_time(start_time=start.time(), end_time=end.time()).plot()
ts60.between_time(start_time=start.time(), end_time=end.time()).plot()
ts60r.between_time(start_time=start.time(), end_time=end.time()).plot()

plt.subplot(224)
plt.title('3 ps from 00:05 on')
ts[pd.datetime(2017,6,19,0,5):].plot()
ts60[pd.datetime(2017,6,19,0,5):].plot()
ts60r[pd.datetime(2017,6,19,0,5):].plot()
plt.tight_layout(pad=0.4, w_pad=0.5, h_pad=1.0) #plt.subplots_adjust(top=0.92, ...)

plt.show()
```



In [33]:
```
#fig, ((ax1, ax2), (ax3, ax4)) = plt.subplots(nrows=2, ncols=2)
#example_plot(ax1)
#def example_plot(ax, fontsize=12):
#    ax.plot(...)
```

"Hand-made" xaxis

In [34]:
```
plt.close('all')
plt.figure()
plt.subplot()
plt.gca().xaxis.set_major_locator(dates.MinuteLocator(interval=10))
plt.gca().xaxis.set_major_formatter(dates.DateFormatter('%H:%M'))
#plt.gcf().autofmt_xdate()
plt.plot(ts.index, ts)
labels=plt.gca().get_xticklabels()
plt.setp(labels,rotation=90)
plt.grid()
plt.show()
```



H5

In [ ]:
```
mx.wrtResultsToH5File()
```

In [36]:
```
mx2=Mx.Mx(h5File='./testdata/M-1-0-1.h5')
```

In [37]:
```
df=mx2.pdf
type(df)
```

Out[37]: pandas.core.frame.DataFrame